2026 1st April

derivative

Let $f(x) = \frac{4x^3+x^2}{x+1}$, find $f'(3)$

$$\frac{(12x^2+2x)(x+1)-(4x^3+x^2)(1)}{(x+1)^2} = \frac{12x^3+12x^2+2x^2+2x-4x^3-x^2}{x^2+2x+1} = \frac{8x^3+13x^2+2x}{x^2+2x+1} =$$

$$= x\frac{8x^2+13x+2}{x^2+2x+1}\ |_{x=3} = \frac{339}{16}$$

Let $f(x) = e^{\sin(2x)}$, find $f'''(0)$

$$f' = e^{\sin(2x)} \cdot 2\cos(2x)$$

$$f'' = \left(e^{\sin(2x)} \cdot 2\cos(2x)\right) \cdot 2\cos(2x) + e^{\sin(2x)} \cdot (-2\sin(2x)) =$$

$$= 4e^{\sin(2x)}\cos^2(2x) - 4e^{\sin(2x)}\sin(2x) = 4e^{\sin(2x)}\left(\cos^2(2x) - \sin(2x)\right)$$

$$f''' = 4e^{\sin(2x)} \cdot 2\cos(2x)\left(\cos^2(2x) - \sin(2x)\right) + 4e^{\sin(2x)}[(2\cos(2x))(-\sin(2x)) - 2\cos(2x)]|_{x=0} = 0$$

integral

$\int_{\ln 6}^{\ln 7} \frac{5e^{3u}+e^{2u}+5e^u+1}{\cosh(u)}\,\mathrm{d}u$

$$x = e^u, \quad u = \ln x, \quad \mathrm{d}u = \frac{1}{x}$$

$$\int_{e^{\ln 6}}^{e^{\ln 7}} \frac{5x^3+x^2+5x+1}{\cosh(\ln x)} \cdot \frac{1}{x}\,\mathrm{d}x,$$

$$\cosh(u) = \frac{e^u+e^{-u}}{2} \Longrightarrow \cosh(\ln(x)) = \frac{e^{\ln x}+e^{-\ln x}}{2} = \frac{x+\frac{1}{x}}{2},$$

$$\int_{e^{\ln 6}}^{e^{\ln 7}} \frac{5x^3+x^2+5x+1}{\frac{x^2+1}{2}}\,\mathrm{d}x = 2\int_{e^{\ln 6}}^{e^{\ln 7}} \frac{5x^3+x^2+5x+1}{x^2+1}\,\mathrm{d}x = 2\int_{e^{\ln 6}}^{e^{\ln 7}} \frac{(x^2+1)(5x+1)}{x^2+1}\,\mathrm{d}x = 2\int_{e^{\ln 6}}^{e^{\ln 7}} 5x+1\,\mathrm{d}x =$$

$$= 2\left[5\frac{x^2}{2}+x\right]_{e^{\ln 6}}^{e^{\ln 7}} = 2\left(\frac{245}{2}+7-90-6\right) = 67$$

2026 2nd April

derivative

Let $f(x) = \sqrt{2x^2+1}$, find $f'(2)$

$$\sqrt{2x^2+1} = (2x^2+1)^{\frac{1}{2}} \overset{()'}{\Longrightarrow} \frac{1}{2}(2x^2+1)^{-\frac{1}{2}}(4x) = \frac{4x}{2\sqrt{2x^2+1}} = \frac{2x}{\sqrt{2x^2+1}}$$

$$f'(2) = \frac{2 \cdot 2}{\sqrt{2 \cdot 4 + 1}} = \frac{4}{3}$$

Let $f(x) = e^{2x}\sec(3x)$, find $f'(\frac{\pi}{12})$

$$\left(e^{2x}\sec(3x)\right)' = 2e^{2x}\sec(3x) + e^{2x}3\tan(3x)\sec(3x) = e^{2x}(2\sec(3x) + 3\tan(3x)\sec(3x)) = e^{2x}(\sec(3x)(2+3\tan(3x)))$$

$$f'\left(\frac{\pi}{12}\right) = e^{\frac{\pi}{6}}\left(\sec\left(\frac{\pi}{4}\right)\left(2+3\tan\left(\frac{\pi}{4}\right)\right)\right)$$

integral

$\int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos^{11}(x)+\cos(x)}\,\mathrm{d}x$

$$\int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos^{10}(x)\cos(x)+\cos(x)}\,\mathrm{d}x = \int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos(x)(\cos^{10}(x)+1)} = \int_0^{\frac{\pi}{4}} \frac{\tan(x)}{\cos^{10}(x)+1} \quad \text{hell no}$$

$$\int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos^{11}(x)+\cos(x)}\,\mathrm{d}x, \quad u = \cos(x), \quad \mathrm{d}u = -\sin(x)\,\mathrm{d}x$$

$$\int \frac{\sin(x)}{u^{11}+u}(-\sin(x))\,\mathrm{d}u = -\int \frac{1}{u^{11}+u}\,\mathrm{d}u = -\left(\int \frac{1}{u^{11}}\,\mathrm{d}u + \int \frac{1}{u}\,\mathrm{d}u\right) = -\left[\frac{u^{-10}}{-10} + \ln(u)\right],$$

$$\left[\frac{1}{10u^{10}} - \ln(u)\right] \Longrightarrow \left[\frac{1}{10\cos^{10}(x) - \ln(\cos(x))}\right]_0^{\frac{\pi}{4}} \text{ dam i messed up bigtime}$$

2026 3rd April

derivative

Let $g(t) = 4e^{t^2-3t+2}$, find $g'(1)$

$$\left(4e^{t^2-3t+2}\right)' = 4(2t-3)e^{t^2-3t+2}$$
$$g'(t) = -4$$

Let $f(x) = \frac{e^{x^2-4}}{\cos(\pi x)}$, find $f'(2)$

$$\left(\frac{e^{x^2-4}}{\cos(\pi x)}\right)' = \frac{2xe^{x^2-4}\cos(\pi x) + \sin(\pi x)e^{x^2-4}}{\cos^2(\pi x)} = \frac{e^{x^2-4}(2x\cos(\pi x) + \sin(\pi x))}{\cos^2(\pi x)}$$
$$f'(2) = \frac{1(4\cdot 1 + 0)}{1^2} = 4$$

Let $f(x) = 5e^{\arcsin(\cos(2x))}$, find $f'(\frac{\pi}{12})$

$$\left(5e^{\arcsin(\cos(2x))}\right)' = 5e^{\arcsin(\cos(2x))}\left(-\frac{\sin(x)}{\sqrt{1-\cos^2(2x)}}\right)$$
$$f'\left(\frac{\pi}{12}\right) = 5e^{\frac{\pi}{3}}\left(-\frac{\frac{\sqrt{6}-\sqrt{2}}{4}}{\sqrt{1-\frac{3}{4}}}\right)$$

well i guess i fucked up

integral

$\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \sin(2\theta)(5\cot(\theta) + 3\tan(\theta))\,\mathrm{d}\theta$

$$\cot = \frac{1}{\tan},\ \ \tan = \frac{\sin}{\cos}, \Longrightarrow \cot = \frac{\cos}{\sin},$$
$$\sin(2\theta)(5\cot(\theta) + 3\tan(\theta)) = 2\sin(\theta)\cos(\theta)\left(5\frac{\cos(\theta)}{\sin(\theta)} + 3\frac{\sin(\theta)}{\cos(\theta)}\right) = 10\cos^2(\theta) + 6\sin^2(\theta) =$$
$$= 10\frac{1+\cos(2\theta)}{2} + 6\frac{1-\cos(2\theta)}{2},$$
$$5\left(\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} 1\,\mathrm{d}\theta + \int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \cos(2\theta)\,\mathrm{d}\theta\right) + 3\left(\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} 1\,\mathrm{d}\theta - \int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \cos(2\theta)\,\mathrm{d}\theta\right) = \frac{2\pi}{3} + 2\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \cos(2\theta)\,\mathrm{d}\theta = \frac{2\pi}{3} + 2\left[\frac{\sin(2\theta)}{2}\right]_{\frac{\pi}{12}}^{\frac{\pi}{6}} =$$
$$= \frac{2\pi}{3} + [\sin(2\theta)]_{\frac{\pi}{12}}^{\frac{\pi}{6}} = \frac{2\pi}{3} + \left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right) = \frac{2\pi}{3} + \frac{\sqrt{3}-1}{2}$$

$\int_0^{\frac{3\pi}{4}} \ln\left(1+\sqrt{\sin(x)+\cos(x)}\right)(\sin(x)-\cos(x))\,\mathrm{d}x$

$$u=\sin(x)+\cos(x),\quad \frac{\mathrm{d}u}{\mathrm{d}x}=\cos(x)-\sin(x),\quad \mathrm{d}x=\frac{\mathrm{d}u}{\cos(x)-\sin(x)}$$

$$\int_1^0 \ln\left(1+\sqrt{u}\right)(\sin(x)-\cos(x))\frac{1}{\cos(x)-\sin(x)}\,\mathrm{d}u=\int_1^0 \ln\left(1+\sqrt{u}\right)\frac{-(\cos(x)-\sin(x))}{\cos(x)-\sin(x)}\,\mathrm{d}u=\int_1^0 -\ln\left(1+\sqrt{u}\right)\mathrm{d}u,$$

$$u=w^2,\quad \frac{\mathrm{d}u}{\mathrm{d}w}=2w,\quad \mathrm{d}u=2w\,\mathrm{d}w,$$

$$\int_1^0 -\ln\left(1+\sqrt{w^2}\right)2w\,\mathrm{d}w=-2\int_1^0 \ln(1+w)w\,\mathrm{d}w,$$

	D	I
$+$	$\ln(1+w)$	w
$-$	$\frac{1}{1+w}$	$\frac{w^2}{2}$

$$-2\left(\frac{\ln(1+w)w^2}{2}-\frac{1}{2}\int_1^0 \frac{w^2}{1+w}\,\mathrm{d}w\right)=-w^2\ln(1+w)+\int_1^0 \frac{w^2}{1+w}\,\mathrm{d}w,$$

$$\int_1^0 \frac{w^2}{1+w}\,\mathrm{d}w=\int_1^0 \frac{w^2-1+1}{1+w}\,\mathrm{d}w=\int_1^0\left(\frac{w^2-1}{w+1}+\frac{1}{w+1}\right)\mathrm{d}w=\int_1^0\left(\frac{(w-1)(w+1)}{w+1}+\frac{1}{w+1}\right)\mathrm{d}w=$$

$$=\int_1^0 w\,\mathrm{d}w-\int_1^0 1\,\mathrm{d}w+\int_1^0 \frac{1}{w+1}\,\mathrm{d}w$$

$$\left[-w^2\ln(1+w)+\frac{1}{2}w^2-w+\ln|w+1|\right]_1^0=0-\left(-\ln(2)+\frac{1}{2}-1+\ln(2)\right)=\frac{1}{2}$$

limit

$\lim_{n\to\infty}\sum_{r=2n+1}^{3n}\sum_{k=-6n}^{6n}\frac{1}{r+6kn}$

$$\text{Mittag-Leffler expansion: } \pi\cot(\pi x)=\lim_{N\to\infty}\sum_{k=-N}^{N}\frac{1}{x+k}$$

$$\sum_{k=-6n}^{6n}\frac{1}{r+6kn},=\frac{1}{6n}\sum_{k=-6n}^{6n}\frac{1}{\frac{r}{6n}+k}\overset{n\to\infty}{\longrightarrow}\frac{1}{6n}\pi\cot\left(\pi\frac{r}{6n}\right),$$

$$\lim_{n\to\infty}\sum_{r=2n+1}^{3n}\frac{1}{6n}\pi\cot\left(\pi\frac{r}{6n}\right)=\int_{\frac{1}{3}}^{\frac{1}{2}}\pi\cot(\pi x)\,\mathrm{d}x,$$

$$u=\pi x,\quad \frac{\mathrm{d}u}{\mathrm{d}x}=\pi,\quad \mathrm{d}x=\frac{\mathrm{d}u}{\pi},$$

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\pi\cot(u)\frac{1}{\pi}\,\mathrm{d}u=\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\cot(u)\,\mathrm{d}u=\left[\ln|\sin(u)|\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}=-\ln\left(\frac{\sqrt{3}}{2}\right)=\ln\left(\frac{2}{\sqrt{3}}\right)$$

2026 4th April

derivative

Let $f(x) = \arctan\left(\frac{1-x^2}{2x}\right)$**, find** $f'\left(\frac{1}{2}\right)$

$$f'(x) = \frac{1}{1+\left(\frac{1-x^2}{2x}\right)^2}\underbrace{\left(\frac{-2x2x-(1-x^2)2}{(2x)^2}\right)}_{\frac{-x^2-1}{2x^2}} = \frac{\frac{-x^2-1}{2x^2}}{1+\left(\frac{1-x^2}{2x}\right)^2}$$

$$f'\left(\frac{1}{2}\right) = \frac{\frac{-\frac{1}{4}-1}{\frac{1}{2}}}{1+\left(\frac{1-\frac{1}{4}}{1}\right)^2} = \frac{-\frac{5}{2}}{\frac{25}{16}} = -\frac{5}{2}\frac{16}{25} = -\frac{80}{50} = -\frac{8}{5}$$

Let $f(x) = \ln(1+\sin(x)-x)$**, find** $f'(0)$

$$f'(x) = \frac{\cos(x)-1}{1+\sin(x)-x}$$

$$f'(0) = \frac{0}{1+1-0} = 0$$

integral

$\int_1^{e^\pi} \cos(\ln(x))\,\mathrm{d}x$

$$u = \ln(x), \quad \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} \Longrightarrow \mathrm{d}x = x\,\mathrm{d}u, \quad x = e^u$$

$$\int_0^\pi \cos(u)e^u\,\mathrm{d}u,$$

	D	I
$+$	e^u	$\cos(u)$
$-$	e^u	$\sin(u)$
$+$	e^u	$-\cos(u)$

$$e^u\sin(u) + e^u\cos(u) - \int_0^\pi \cos(u)e^u\,\mathrm{d}u,$$

$$2I = e^u(\sin(u)+\cos(u))$$

$$I = \frac{e^u(\sin(u)+\cos(u))}{2}$$

$$\left[\frac{e^u(\sin(u)+\cos(u))}{2}\right]_0^\pi = -\frac{e^\pi}{2} - \frac{1}{2}$$

$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (2x + 2x\cot^2(x) + 2x\tan^2(x) + 2x\tan^2(x)\cot^2(x))\,\mathrm{d}x$

$$2x + 2x\cot^2(x) + 2x\tan^2(x) + 2x\tan^2(x)\cot^2(x) = 2x(\cot^2(x) + \tan^2(x) + 2) =$$
$$= 2x((1 + \cot^2(x)) + (1 + \tan^2(x))) = 2x(\csc^2(x) + \sec^2(x))$$
$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} 2x(\csc^2(x) + \sec^2(x))\,\mathrm{d}x$$

	D	I
$+$	$2x$	$\csc^2(x) + \sec^2(x)$
$-$	2	$\tan(x) - \cot(x)$
$+$	0	$-\ln\lvert\cos(x)\rvert - \ln\lvert\sin(x)\rvert$

$$[2x(\tan(x) - \cot(x)) - 2(-\ln|\cos(x)| - \ln|\sin(x)|)]_{\frac{\pi}{6}}^{\frac{\pi}{4}} = [2(x(\tan(x) - \cot(x)) + \ln|\cos(x)| + \ln|\sin(x)|)]_{\frac{\pi}{6}}^{\frac{\pi}{4}} =$$
$$= \frac{2\pi\sqrt{3}}{9} + \ln\left(\frac{4}{3}\right)$$

limit

$\lim_{n\to\infty} \sum_{m=0}^{n} \frac{1}{\int_0^{2m}\{x\}\,\mathrm{d}x}$

$$\int_0^{2m} \{x\}\,\mathrm{d}x = 2^m \cdot \frac{1}{2} = 2^{m-1},$$
$$\lim_{n\to\infty} \sum_{m=0}^{n} \frac{1}{2^{m-1}} = \frac{2}{1 - \frac{1}{2}} = 4$$

2026 5th April

derivative

easy

trivial, $f'(1) = \frac{1}{2\sqrt{2}}$

medium

Let $f(x) = \sqrt{\ln\left(\sqrt{\ln(\sqrt{x})}\right)}$, find $f'(8)$

$$a = \sqrt{b}, \quad b = \ln(c), \quad c = \sqrt{d}, \quad d = \ln(e), \quad e = \sqrt{x},$$

$$a' = \frac{1}{2\sqrt{b}}, \quad b' = \frac{1}{c}, \quad c' = \frac{1}{2\sqrt{d}}, \quad d' = \frac{1}{e}, \quad e' = \frac{1}{2\sqrt{x}},$$

$$f'(x) = \frac{1}{2\sqrt{b}} \cdot \frac{1}{c} \cdot \frac{1}{2\sqrt{d}} \cdot \frac{1}{e} \cdot \frac{1}{2\sqrt{x}},$$

$$f'(8) = \frac{1}{96\ln(2)\sqrt{\ln\left(\sqrt{\frac{3}{2}\ln(2)}\right)}}$$

integral

beginner

Let $f(x) = \int_4^{12} \sqrt{8x+4}e^{\sqrt{2x+1}}\,\mathrm{d}x$

$$\int_4^{12} \sqrt{8x+4}e^{\sqrt{2x+1}}\,\mathrm{d}x = \int_4^{12} \sqrt{4(2x+1)}e^{\sqrt{2x+1}}\,\mathrm{d}x = \int_4^{12} \sqrt{4}\sqrt{2x+1}e^{\sqrt{2x+1}}\,\mathrm{d}x,$$

$$u = \sqrt{2x+1}, \quad \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2\sqrt{2x+1}} \cdot 2 \Longrightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u}, \quad \mathrm{d}x = u\,\mathrm{d}u,$$

$$\int_3^5 2ue^u u\,\mathrm{d}u = 2\int_3^5 u^2 e^u\,\mathrm{d}u$$

	D	I
$+$	u^2	e^u
$-$	$2u$	e^u
$+$	2	e^u
$-$	0	e^u

$$2\left[e^u(u^2-2u+2)\right]_3^5 = 2\left[17e^5 - 5e^3\right]$$

easy

Let $\int_0^1 x\ln(x^2+1)\,\mathrm{d}x$

$$u = x^2+1, \quad \frac{\mathrm{d}u}{\mathrm{d}x} = 2x, \quad \mathrm{d}x = \frac{\mathrm{d}u}{2x}, \quad x = \sqrt{u-1},$$

$$\int_1^2 \sqrt{u-1}\ln(u)\frac{1}{2\sqrt{u-1}}\,\mathrm{d}u = \frac{1}{2}\int_1^2 \ln(u)\,\mathrm{d}u = \frac{1}{2}[u\ln(u)-u]_1^2 = \frac{1}{2}[(2\ln(2)-2)+1] = \ln(2) - \frac{1}{2}$$

limit

Let $f(x) = \frac{\arctan(x)}{x}$, find $\lim_{x\to\infty} f^{(4)}(x)$

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - ...$$

$$f(x) = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \frac{x^6}{7} + \frac{x^8}{9} - ...$$

$$f^{(4)}(x) = 0 - 0 + \frac{24}{5} -?\, x^2 +?\, x^4$$

2026 6th April

derivative

easy

Let $f(x) = x^3 + x^2 f'(1) + 2x f''(2) + f'''(3), \quad x \in \mathbb{R}, \quad$ find $f'(5)$

$$A = f'(1), \quad B = f''(2), \quad C = f'''(3),$$

$$f(x) = x^3 + Ax^2 + 2Bx + C, \quad f'(x) = 3x^2 + 2xA + 2B, \quad f''(x) = 6x + 2A, \quad f'''(x) = 6,$$

$$f'''(3) = 6 = C,$$

$$f''(2) = 12 + 2A = B,$$

$$f'(1) = 3 + 2A + 2B = A \Longrightarrow A = 3 + 2A + 2(12 + 2A) \Longrightarrow A = 27 + 6A \Longrightarrow A = -\frac{27}{5},$$

$$B = 12 + 2\left(-\frac{27}{5}\right) = \frac{60}{5} - \frac{54}{5} \Longrightarrow B = \frac{6}{5},$$

$$f'(5) = 75 + 10\left(-\frac{27}{5}\right) + \frac{12}{5} = 75 - 54 + \frac{12}{5} = 21 + \frac{12}{5} = \frac{117}{5}$$

integral

easy

$\int_0^1 (2x^3 - 3x^2 - x + 1)^{\frac{1}{3}} \, \mathrm{d}x$

$$\int_0^1 (2x^3 - 3x^2 - x + 1)^{\frac{1}{3}} \, \mathrm{d}x = \left[\frac{(2x^3 - 3x^2 - x + 1)^{\frac{4}{3}}}{\frac{4}{3}}\right]_0^1 = -\frac{3}{4} + \frac{3}{4} = 0$$

limit

2026 7th April

derivative

easy

Let $f(x) = \int_0^x \frac{\sin(t)}{1+t^2}\,\mathrm{d}t + \int_x^1 \frac{\cos(t)}{1+t^2}\,\mathrm{d}t$, find $f'(1)$

$$\left(\int_0^x \frac{\sin(t)}{1+t^2}\,\mathrm{d}t\right)' = \frac{\sin(x)}{1+x^2} - 0 = \frac{\sin(x)}{1+x^2},$$

$$\left(\int_x^1 \frac{\cos(t)}{1+t^2}\,\mathrm{d}t\right)' = 1 - \frac{\cos(x)}{1+x^2} = -\frac{\cos(x)}{1+x^2},$$

$$f'(x) = \frac{\sin(x) - \cos(x)}{1+x^2}, \qquad f'(1) = \frac{\sin(1) - \cos(1)}{2}$$

medium

Let $f(x) = \frac{e^x \sin(x)}{1+x} + \left(e - 1 - \frac{1}{10000}\right)x$, find $f'(0)$

$$f'(x) = \frac{(e^x \sin(x) + e^x \cos(x))(1+x) - (e^x \sin(x))}{(1+x)^2} + e - 1 - \frac{1}{10000},$$

$$f'(1) = \frac{1-0}{1} + e - 1 - \frac{1}{10000} = e - \frac{1}{10000}$$

integral

beginner

$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\sin^2(x)\cos^2(x)}\,\mathrm{d}x$

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\sin^2(x)\cos^2(x)}\,\mathrm{d}x = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin^2(x) + \cos^2(x)}{\sin^2(x)\cos^2(x)}\,\mathrm{d}x = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin^2(x)}{\sin^2(x)\cos^2(x)}\,\mathrm{d}x + \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\cos^2(x)}{\sin^2(x)\cos^2(x)}\,\mathrm{d}x =$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\cos^2(x)}\,\mathrm{d}x + \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\sin^2(x)}\,\mathrm{d}x = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \csc^2(x)\,\mathrm{d}x + \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sec^2(x)\,\mathrm{d}x = [\tan(x) - \cot(x)]_{\frac{\pi}{6}}^{\frac{\pi}{4}} =$$

$$= \left(1 - 1 - \left(\frac{\sqrt{3}}{2} - \sqrt{3}\right)\right) = \frac{2\sqrt{3}}{3}$$

easy

$\int_0^1 \left[x^x(\ln x + 1) + \cos(x^2) \cdot 2x + \frac{1}{1+x^2}\right] \mathrm{d}x$

$$\underbrace{\int_0^1 x^x(\ln x + 1)\,\mathrm{d}x}_{A} + \underbrace{\int_0^1 \cos(x^2) \cdot 2x\,\mathrm{d}x}_{B} + \underbrace{\int_0^1 \frac{1}{1+x^2}\,\mathrm{d}x}_{C},$$

$$y = x^x, \;\; \ln(y) = \ln(x^x), \;\; \ln(y) = x\ln(x),$$

$$\frac{y'}{y} = x'\ln(x) + x\frac{1}{x}, \;\; \frac{y'}{y} = \ln(x) + 1, \;\; y' = y\ln(x) + 1, \;\; y' = x^x(\ln(x) + 1), \Longrightarrow A = [x^x]_0^1$$

$$u = x^2, \;\; \frac{\mathrm{d}u}{\mathrm{d}x} = 2x, \;\; \mathrm{d}x = \frac{\mathrm{d}u}{2x},$$

$$\int_0^1 \cos(u)\,\mathrm{d}u = [\sin(u)]_0^1 \Longrightarrow B = \left[\sin(x^2)\right]_0^1$$

$$\int_0^1 \frac{1}{1+x^2}\,\mathrm{d}x = [\arctan(x)]_0^1 = C,$$

$$I = \left[x^x + \sin(x^2) + \arctan(x)\right]_0^1 = (1 + \sin(1) + \arctan(1)) - (0 + 1 + 0) = \sin(1) + \arctan(1)$$

2026 8th April

derivative

easy

Let $y(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$, find $y'(-1) - y''(-1)$

$$(1-x)y(x) = (1+x)(1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16}) \Longrightarrow (1-x)y(x) = 1-x^{32} \Longrightarrow y(x) = \frac{1-x^{32}}{1-x}$$

$$y'(x) = \frac{-32x^{31}(1-x) + (1-x^{32})}{(1-x)^2}, \quad y'(-1) = \frac{64}{4} = 16$$

$$y'(x) = \frac{-32x^{31} + 32x^{32} + 1 - x^{32}}{(1-x)^2},$$

$$y''(x) = \frac{(-992x^{30} + 1024x^{31} - 32x^{31})(1-x)^2 + (-32x^{31} + 32x^{32} + 1 - x^{32})(2-2x)}{(1-x)^4},$$

$$y''(-1) = \frac{(-992 - 1024 + 32)4 + (32 + 32 + 1 - 1)4}{16} = \frac{-1984 + 64}{4} = -\frac{1920}{4} = -480,$$

$$16 - (-480) = 496$$

medium

Let $f(x) = \ln\left(x + \sqrt{x^2+4}\right) - \frac{x}{\sqrt{x^2+4}}$, find $f'(2)$

$$f'(x) = \frac{1 + \frac{x}{\sqrt{x^2+4}}}{x + \sqrt{x^2+4}} - \frac{\sqrt{x^2+4} - \frac{x^2}{\sqrt{x^2+4}}}{x^2+4}$$

$$f'(2) = \frac{1 + \frac{2}{\sqrt{8}}}{2 + \sqrt{8}} - \frac{\sqrt{8} - \frac{4}{\sqrt{8}}}{8}$$

integral

beginner

$\int_0^1 \frac{\arcsin(x)}{\sqrt{1-x^2}}\,\mathrm{d}x$

$$\int_0^1 \arcsin(x)\frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x = \int_0^1 \arcsin(x)(\arcsin(x))'\,\mathrm{d}x,$$

$$u = \arcsin(x), \quad \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}}, \quad \mathrm{d}x = \sqrt{1-x^2}\,\mathrm{d}u,$$

$$\int_0^{\frac{\pi}{2}} u\,\mathrm{d}u = \left[\frac{1}{2}u^2\right]_0^{\frac{\pi}{2}} = \frac{\pi^2}{8}$$

easy

$\int_0^1 \left(\frac{x\sqrt{\arctan(x)}}{\sqrt{1+x^2}}\right)^2 \mathrm{d}x$

$$\int_0^1 \frac{x^2 \arctan(x)}{1+x^2}\,\mathrm{d}x$$

$$u = \arctan(x), \quad \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2}, \quad \mathrm{d}x = (1+x^2)\,\mathrm{d}u, \quad x = \tan(u)$$

$$\int_0^{\frac{\pi}{4}} \tan^2(u)u\,\mathrm{d}u = \int_0^{\frac{\pi}{4}} u(\sec^2(u)-1)\,\mathrm{d}u = \int_0^{\frac{\pi}{4}} u\sec^2(u)\,\mathrm{d}u - \int_0^{\frac{\pi}{4}} u\,\mathrm{d}u,$$

	D	I
$+$	u	$\sec^2(u)$
$-$	1	$\tan(u)$
$+$	0	$-\ln\lvert\cos(u)\rvert$

$$\left[u\tan(u) + \ln|\cos(u)| - \frac{1}{2}u^2\right]_0^{\frac{\pi}{4}} = \left(\frac{\pi}{4}(1) + \ln\left(\frac{\sqrt{2}}{2}\right) - \frac{1}{2}\left(\frac{\pi}{4}\right)^2\right) - (0 + \ln(1) - 0) = \frac{\pi}{4} - \frac{1}{2}\ln(2) - \frac{\pi^2}{32}$$

2026 9th April

derivative

easy

Let $f(x) = \ln(x^2 + e^x) + \arctan(\sqrt{x})$, find $g'(f(1))$, where g is the inverse of f

$$g(f(x)) = x,$$
$$g'(f(x)) \cdot f'(x) = 1,$$
$$g'(f(x)) = \frac{1}{f'(x)} \Longrightarrow g'(f(1)) = \frac{1}{f'(1)},$$
$$f'(x) = \frac{2x + e^x}{x^2 + e^x} + \frac{1}{(1+x)2\sqrt{x}} = \frac{2x + e^x}{x^2 + e^x} + \frac{1}{2\sqrt{x} + 2x},$$
$$f'(1) = \frac{2+e}{1+e} + \frac{1}{4} \Longrightarrow \frac{9+5e}{4+4e}$$
$$\frac{1}{f'(1)} = \frac{4+4e}{9+5e}$$

medium

Let $f(x) = \cos(\cos(\cos(...(x)...)))$, find $f'(\pi)$

$$y = \cos(\cos(\cos(...(x)...))) \Longrightarrow y = \cos(y),$$
$$y' = -\sin(y) \cdot y',$$
$$y' + \sin(y) \cdot y' = 0,$$
$$y' \underbrace{(1 + \sin(y))}_{\neq 0} = 0 \Longrightarrow y' = 0$$

integral

beginner

$\int_0^1 \frac{x \ln(x^2+1)}{x^2+1} \, \mathrm{d}x$

$$u = x^2 + 1, \quad \frac{\mathrm{d}u}{\mathrm{d}x} = 2x, \quad \mathrm{d}x = \frac{\mathrm{d}u}{2x}$$
$$\int_1^2 \frac{x \ln(u)}{u} \cdot \frac{1}{2x} \, \mathrm{d}u = \int_1^2 \frac{\ln(u)}{2u} \, \mathrm{d}u = \frac{1}{2} \int_1^2 \ln(u) \cdot \frac{1}{u} \, \mathrm{d}u = \frac{1}{2} \left[\frac{1}{2} \ln^2(u) \right]_1^2 = \frac{1}{4} \ln^2(2)$$

easy

$\int_1^\infty \frac{264}{x(x^4+66)}\,\mathrm{d}x$

$$264\int_1^\infty \frac{1}{x(x^4+66)}\,\mathrm{d}x = 264\int_1^\infty \frac{x^3}{x^4(x^4+66)}\,\mathrm{d}x,$$

$$u = x^4,\quad \frac{\mathrm{d}u}{\mathrm{d}x} = 4x^3,\quad \mathrm{d}x = \frac{1}{4x^3}\,\mathrm{d}u,$$

$$264\lim_{t\to\infty}\int_1^t \frac{x^3}{u(u+66)}\cdot\frac{1}{4x^3}\,\mathrm{d}u = \frac{264}{4}\lim_{t\to\infty}\int_1^t \frac{1}{u(u+66)}\,\mathrm{d}u,$$

$$\frac{A}{u}+\frac{B}{u+66} = 1 \Longrightarrow Au+66A+Bu = 1 \Longrightarrow u(A+B)+66A = 1 \Longrightarrow A = \frac{1}{66},\quad B = -\frac{1}{66},$$

$$\frac{264}{4}\lim_{t\to\infty}\left(\int_1^t \frac{\frac{1}{66}}{u}\,\mathrm{d}u - \int_1^t \frac{\frac{1}{66}}{u+66}\,\mathrm{d}u\right) = \lim_{t\to\infty}\left[\ln|u| - \ln|u+66|\right]_1^t = \lim_{t\to\infty}\left[\ln\left|\frac{u}{u+66}\right|\right]_1^t = -\ln\left(\frac{1}{67}\right)$$

2026 10th April

shit i forgor xdd

2026 11th April

derivative

easy

Let $f(x) = \frac{\sqrt{1+\sin(x^2)}}{1+\cos(x^2)}$, find $f'(1)$

$$f(x) = \frac{\sqrt{1+\sin(x^2)}}{1+\cos(x^2)},$$

$$u(x) = \sqrt{1+\sin(x^2)} \Longrightarrow u'(x) = \frac{x\cos(x^2)}{\sqrt{1+\sin(x^2)}},$$
$$v(x) = 1+\cos(x^2) \Longrightarrow v'(x) = -2x\sin(x^2),$$

$$f'(x) = \frac{u'(x)v(x) - u(x)v'(x)}{v(x)^2} = \frac{\frac{x\cos(x^2)}{\sqrt{1+\sin(x^2)}}(1+\cos(x^2)) - \left(\sqrt{1+\sin(x^2)}\right)(-2x\sin(x^2))}{(1+\cos(x^2))^2},$$

$$f'(1) = \frac{\frac{\cos(1)}{\sqrt{1+\sin(1)}}(1+\cos(1)) - \left(\sqrt{1+\sin(1)}\right)(-2\sin(1))}{(1+\cos(1))^2},$$

$$f'(1) = \frac{\cos(1)(1+\cos(1)) + 2\sin(1)(1+\sin(1))}{\sqrt{1+\sin(1)}(1+\cos(1))^2},$$

$$f'(1) = \frac{\cos(1) + \cos^2(1) + 2\sin(1) + 2\sin^2(1)}{\sqrt{1+\sin(1)}(1+\cos(1))^2},$$

$$f'(1) = \frac{\cos(1) + 2\sin(1) + 1 + \sin^2(1)}{\sqrt{1+\sin(1)}(1+\cos(1))^2}$$

medium

Let $f(x) = \arctan\left(\sqrt{\frac{1-x}{1+x}}\right)$, find $f'\left(\frac{1}{2}\right)$

$$\left(\frac{1-x}{1+x}\right)' = \frac{-1(1+x)-(1-x)}{(1+x)^2} = \frac{-1-x-1+x}{(1+x)^2} = -\frac{2}{(1+x)^2}$$

$$\left(\sqrt{\frac{1-x}{1+x}}\right)' = \frac{1}{2\sqrt{\frac{1-x}{1+x}}}\left(-\frac{2}{(1+x)^2}\right) = -\frac{1}{\sqrt{\frac{1-x}{1+x}}(1+x)^2} = -\frac{\sqrt{1+x}}{\sqrt{1-x}(1+x)^2}$$

$$f'(x) = \frac{1}{1+\left(\sqrt{\frac{1-x}{1+x}}\right)^2}\left(-\frac{\sqrt{1+x}}{\sqrt{1-x}(1+x)^2}\right) = \frac{1+x}{2}\left(-\frac{\sqrt{1+x}}{\sqrt{1-x}(1+x)^2}\right) = -\frac{1}{2\sqrt{1-x}\sqrt{1+x}}$$

$$f'\left(\frac{1}{2}\right) = -\frac{1}{2\sqrt{\frac{1}{2}}\sqrt{\frac{3}{2}}}$$

integral

beginner

$\int_0^1 \frac{1}{\sqrt{1+x^2}+x}\,\mathrm{d}x$

$$u=\sqrt{1+x^2},\quad \frac{\mathrm{d}u}{\mathrm{d}x}=\frac{2x}{2\sqrt{1+x^2}}=\frac{x}{\sqrt{1+x^2}},\quad \mathrm{d}x=\frac{\sqrt{1+x^2}}{x}\,\mathrm{d}u,\quad x=\sqrt{u^2-1},$$

$$\int_1^{\sqrt{2}}\left(\frac{1}{u+\sqrt{u^2-1}}\right)\left(\frac{u}{\sqrt{u^2-1}}\right)\mathrm{d}u=\int_0^{\sqrt{2}}\frac{\sqrt{u^2-1}+u^2\sqrt{u^2-1}}{u\sqrt{u^2-1}(u^2-1)}\,\mathrm{d}u=\int_0^{\sqrt{2}}\frac{1+u^2}{u^3-u}\,\mathrm{d}u,$$

this is NOT the way

$$\int_0^1\left(\frac{1}{\sqrt{1+x^2}+x}\right)\left(\frac{\sqrt{1+x^2}-x}{\sqrt{1+x^2}-x}\right)\mathrm{d}x=\int_0^1\frac{\sqrt{1+x^2}-x}{1+x^2-x^2}\,\mathrm{d}x=\int_0^1\sqrt{1+x^2}-x\,\mathrm{d}x=$$

$$=\int_0^1\sqrt{1+x^2}\,\mathrm{d}x-\int_0^1 x\,\mathrm{d}x=\left[\frac{1}{2}\left(x\sqrt{1+x^2}+\ln\left(x+\sqrt{1+x^2}\right)-x^2\right)\right]_0^1=\frac{1}{2}\left(\sqrt{2}+\ln\left(1+\sqrt{2}\right)-1\right)$$

easy

$\int_0^1 \frac{e^{3u}}{e^u+1}\,\mathrm{d}u$

$$x=e^u,\quad \frac{\mathrm{d}x}{\mathrm{d}u}=e^u,\quad \mathrm{d}u=\frac{\mathrm{d}x}{e^u}$$

$$\int_1^e\left(\frac{x^3}{x+1}\right)\left(\frac{1}{x}\right)\mathrm{d}x=\int_1^e\frac{x^2}{x+1}\,\mathrm{d}x=\int_1^e\frac{x^2+1-1}{x+1}\,\mathrm{d}x=\int_1^e\frac{(x-1)(x+1)+1}{x+1}\,\mathrm{d}x=$$

$$=\int_1^e\frac{(x-1)(x+1)}{x+1}\,\mathrm{d}x+\int_1^e\frac{1}{x+1}\,\mathrm{d}x=\left[\frac{x^2}{2}-x+\ln|x+1|\right]_1^e=\frac{e^2}{2}-e+\ln(e+1)-\left(\frac{1}{2}-1+\ln(1+1)\right)=$$

$$=\frac{e^2}{2}-e+\ln(e+1)+\frac{1}{2}-\ln(2)$$

2026 12th April

derivative

easy

Let $f(x) = x\sqrt{1-x^2} + \arcsin(x)$, find $f'\left(\frac{1}{2}\right)$

$$f'(x) = \sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}},$$

$$f'\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2} - \frac{\frac{1}{4}}{\frac{\sqrt{3}}{2}} + \frac{1}{\frac{\sqrt{3}}{2}} = \frac{\sqrt{3}}{2} + \frac{\frac{3}{4}}{\frac{\sqrt{3}}{2}}$$

medium

Let $f(x) = e^{x\sqrt{x}}$, find $f''(1)$

$$f'(x) = \frac{3\sqrt{x^3}e^{\sqrt{x^3}}}{2x} = \frac{3\sqrt{x}e^{\sqrt{x^3}}}{2},$$

$$f''(x) = \left(\frac{3}{4\sqrt{x}}\right)\left(e^{\sqrt{x^3}}\right) + \left(\frac{3}{2}\sqrt{x}\right)\left(\frac{3}{2}\sqrt{x}e^{\sqrt{x^3}}\right),$$

$$f''(1) = \frac{3}{4}e + \frac{9}{4}e = 3e$$

integral

beginner

$\int_0^{\frac{\pi}{2}} e^{2x}(2\sin(x) + \cos(x))\,\mathrm{d}x$

$$\int_0^{\frac{\pi}{2}} e^{2x}(2\sin(x) + \cos(x))\,\mathrm{d}x = \int_0^{\frac{\pi}{2}} e^{2x}\sin(2x)\,\mathrm{d}x,$$

$$u = 2x, \quad \frac{\mathrm{d}u}{\mathrm{d}x} = 2, \quad \mathrm{d}x = \frac{1}{2}\,\mathrm{d}u,$$

$$\int_0^{\pi} e^u \sin(u)\frac{1}{2}\,\mathrm{d}u = \frac{1}{2}\int_0^{\pi} e^u \sin(u)\,\mathrm{d}u,$$

	D	I
$+$	e^u	$\sin(u)$
$-$	e^u	$-\cos(u)$
$+$	e^u	$-\sin(u)$

$$I = \frac{1}{2}(e^u(-\cos(u)) + e^u(\sin(u)) - I) = \frac{1}{2}e^u(\sin(u) - \cos(u)) - \frac{1}{2}I,$$

$$\frac{3}{2}I = \frac{1}{2}e^u(\sin(u) - \cos(u)),$$

$$I = \frac{1}{3}e^u(\sin(u) - \cos(u)),$$

$$\left[\frac{1}{3}e^u(\sin(u) - \cos(u))\right]_0^{\pi} = \frac{1}{3}e^{\pi} - \frac{1}{3},$$

guess fucking what turns out fucking $2\sin(x) + \cos(x)$ is not $\sin(2x)$ fuck

$$\int_0^{\frac{\pi}{2}} e^{2x}(2\sin(x)+\cos(x))\,\mathrm{d}x = \left[e^{2x}\sin(x)\right]_0^{\frac{\pi}{2}} = e^{\pi}$$

easy

$\int_0^1 \frac{2x}{1+\sqrt{3x^2+1}}\,\mathrm{d}x$

$$u=\sqrt{3x^2+1}, \qquad \frac{\mathrm{d}u}{\mathrm{d}x}=\frac{3x}{\sqrt{3x^2+1}}, \qquad \mathrm{d}x=\frac{\sqrt{3x^2+1}}{3x}\,\mathrm{d}u,$$

$$\int_1^2\left(\frac{2x}{1+u}\right)\left(\frac{u}{3x}\right)\mathrm{d}u=\int_1^2\frac{2u}{3(1+u)}\,\mathrm{d}u=\frac{2}{3}\int_1^2\frac{u}{1+u}\,\mathrm{d}u=\frac{2}{3}\int_1^2\frac{1+u-1}{1+u}\,\mathrm{d}u=$$

$$=\frac{2}{3}\left(\int_1^2 1\,\mathrm{d}u-\int_1^2\frac{1}{1+u}\,\mathrm{d}u\right)=\frac{2}{3}[u-\ln|1+u|]_1^2=\frac{2}{3}(2-\ln(3)-(1-\ln(2)))=\frac{2}{3}\left(1+\ln\left(\frac{2}{3}\right)\right)$$

medium

$\int_{\frac{\sqrt{2}-26}{2}}^{\frac{\sqrt{3}-26}{2}} \arctan\left(\frac{x+13}{\sqrt{-x^2-26x-168}}\right)\mathrm{d}x$

$$\int_{\frac{\sqrt{2}-26}{2}}^{\frac{\sqrt{3}-26}{2}} \arctan\left(\frac{x+13}{\sqrt{-(x+12)(x+14)}}\right)\mathrm{d}x, \text{ hell naw}$$

$$-x^2-26x-168=-x^2-26x-168-1+1=-\left((x+13)^2-1\right)=1-(x+13)^2,$$

$$\int_{\frac{\sqrt{2}-26}{2}}^{\frac{\sqrt{3}-26}{2}} \arctan\left(\frac{x+13}{\sqrt{1-(x+13)^2}}\right)\mathrm{d}x,$$

$$u=x+13, \qquad \frac{\mathrm{d}u}{\mathrm{d}x}=1, \qquad \mathrm{d}x=\mathrm{d}u,$$

$$\frac{\sqrt{2}-26}{2}+\frac{26}{2}=\frac{\sqrt{2}}{2}, \qquad \frac{\sqrt{3}-26}{2}+\frac{26}{2}=\frac{\sqrt{3}}{2}$$

$$\int_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} \arctan\left(\frac{u}{\sqrt{1-u^2}}\right)\mathrm{d}u,$$

$$\sin(\theta)=u \Longrightarrow \theta=\arcsin(u), \qquad \tan(\theta)=\frac{u}{\sqrt{1-u^2}},$$

$$\int_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} \arctan(\tan(\theta))\,\mathrm{d}u=\int_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} \arcsin(\theta)\,\mathrm{d}u=\left[u\arcsin(u)+\sqrt{1-u^2}\right]_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}}=$$

$$=\left(\frac{\sqrt{3}}{2}\frac{\pi}{3}+\frac{1}{2}\right)-\left(\frac{\sqrt{2}}{2}\frac{\pi}{4}+\frac{\sqrt{2}}{2}\right)=\frac{\sqrt{3}\pi}{6}+\frac{1}{2}-\frac{\sqrt{2}\pi}{8}-\frac{\sqrt{2}}{2}=\frac{4\pi\sqrt{3}+12-3\pi\sqrt{2}-12\sqrt{2}}{24}$$

2026 13th April

derivative

easy

Let $f(x) = x^{\frac{\ln(\ln x)}{\ln x}}$, find $f'(e)$

$$f'(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\ln(\ln x)}{\ln x}\right) = \frac{\frac{1}{x\ln x}\ln x - \ln(\ln x)\frac{1}{x}}{\ln^2 x} = \frac{\frac{1}{x} - \frac{\ln(\ln x)}{x}}{\ln^2 x},$$

$$f'(e) = \frac{\frac{1}{e} - \frac{0}{e}}{\ln^2 e} = \frac{\frac{1}{e}}{1} = \frac{1}{e}$$

medium

Let $f(x) = \frac{\sin(2x)\cosh(x) + \ln(1+x)\cos(x)}{1+x^2}$, find $f'(1)$

$$f'(x) =$$

$$= \frac{\left[2\cos(2x)\cosh(x) + \sin(2x)\sinh(x) + \frac{\cos(x)}{1+x} + \ln(1+x)(-\sin(x))\right](1+x^2) - [\sin(2x)\cosh(x) + \ln(1+x)\cos(x)](2x)}{(1+x^2)^2},$$

$$f'(1) = \frac{2\cos(2)\cosh(1) + \sin(2)\sinh(1) + \frac{\cos(1)}{2} - \ln(2)\sin(1) - \sin(2)\cosh(1) - \ln(2)\cos(1)}{2}$$

integral

beginner

$\int_0^1 \frac{\arctan(\sqrt{x})}{\sqrt{x}(1+x)}\,\mathrm{d}x$

$$u = \sqrt{x}, \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2\sqrt{x}}, \qquad \mathrm{d}x = 2\sqrt{x}\,\mathrm{d}u, \qquad x = u^2,$$

$$\int_0^1 \frac{\arctan(u)}{u(1+u^2)} \cdot 2u\,\mathrm{d}u = 2\int_0^1 \frac{\arctan(u)}{1+u^2}\,\mathrm{d}u = 2\int_0^1 \arctan(u) \cdot \frac{1}{1+u^2} = 2\frac{1}{2}\left[\arctan^2(u)\right]_0^1 = \arctan^2(1)$$

easy

$\int_0^1 \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}\,\mathrm{d}x$

$$u = \sqrt{x}, \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2\sqrt{x}}, \qquad \mathrm{d}x = 2\sqrt{x}\,\mathrm{d}u, \qquad x = u^2,$$

$$\int_0^1 \sqrt{\frac{1-u}{1+u}} \cdot 2u\,\mathrm{d}u,$$

$$u = \cos(\theta), \qquad \frac{\mathrm{d}u}{\mathrm{d}\theta} = -\sin(\theta), \qquad \mathrm{d}\theta = -\sin(\theta)\,\mathrm{d}u,$$

$$2\int_0^{\frac{\pi}{2}} \sqrt{\frac{1-\cos(\theta)}{1+\cos(\theta)}}(\cos(\theta))(-\sin(\theta))\,\mathrm{d}\theta,$$

$$\sqrt{\frac{1-\cos(\theta)}{1+\cos(\theta)}} = \sqrt{\frac{(1-\cos(\theta))(1-\cos(\theta))}{(1+\cos(\theta))(1-\cos(\theta))}} = \sqrt{\frac{(1-\cos(\theta))^2}{1-\cos^2(\theta)}} = \frac{1-\cos(\theta)}{\sin(\theta)},$$

$$2\int_0^{\frac{\pi}{2}} \frac{1-\cos(\theta)}{\sin(\theta)}\cos(\theta)\sin(\theta)\,\mathrm{d}\theta = 2\int_0^{\frac{\pi}{2}} \cos(\theta) - \cos^2(\theta)\,\mathrm{d}\theta = 2\left[\sin(\theta) - \left(\frac{\theta}{2} + \frac{\sin(2\theta)}{4}\right)\right]_0^{\frac{\pi}{2}} =$$

$$= 2\left(1 - \frac{\pi}{4}\right) = 2 - \frac{\pi}{2}$$

2026 14th April

derivative

easy

Let $f(x) = x\sin(x)$, find $f''(\frac{\pi}{4})$

$$f'(x) = \sin(x) + x\cos(x),$$
$$f''(x) = \cos(x) + \cos(x) + x(-\sin(x)) = 2\cos(x) - x\sin(x)$$
$$f''\left(\frac{\pi}{4}\right) = 2\frac{\sqrt{2}}{2} - \frac{\pi}{4}\frac{\sqrt{2}}{2} = \sqrt{2} - \frac{\pi\sqrt{2}}{8}$$

medium

Find $\frac{\mathrm{d}^{17}}{\mathrm{d}x^{17}}(e^x\sin(x))_{|x=0}$

$$f'(x) = e^x\cos(x) + e^x\sin(x),$$
$$f''(x) = -e^x\sin(x) + e^x\cos(x) + e^x\cos(x) + e^x\sin(x) = 2e^x\cos(x),$$
$$f'''(x) = -\sin(x)2e^x + 2e^x\cos(x),$$
$$f^{(4)} = -\cos(x)2e^x - \sin(x)2e^x - 2e^x\sin(x) + 2e^x\cos(x) = 4e^x\sin(x),$$
$$\dots$$
$$f^{(16)} = 256e^x\sin(x),$$
$$f^{(17)} = 256e^x\cos(x) + 256e^x\sin(x) = 256e^x(\cos(x) + \sin(x)),$$
$$f^{(17)}(0) = 256$$

integral

beginner

$\int_0^{\frac{\pi}{2}} \cos(x)e^{\sqrt{\sin(x)}}\,\mathrm{d}x$

$$u = \sin(x), \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = \cos(x), \qquad \mathrm{d}x = \frac{1}{\cos(x)}\,\mathrm{d}u,$$
$$\int_0^1 \cos(x)e^{\sqrt{u}}\frac{1}{\cos(x)}\,\mathrm{d}u = \int_0^1 e^{\sqrt{u}}\,\mathrm{d}u,$$
$$v = \sqrt{u}, \qquad \frac{\mathrm{d}v}{\mathrm{d}u} = \frac{1}{2\sqrt{u}}, \qquad \mathrm{d}u = 2\sqrt{u}\,\mathrm{d}v,$$
$$\int_0^1 e^v 2v\,\mathrm{d}v = 2\int_0^1 e^v v\,\mathrm{d}v,$$

	D	I
$+$	v	e^v
$-$	1	e^v
$+$	0	e^v

$$I = e^v v - e^v = e^v(v-1),$$
$$2[e^v(v-1)]_0^1 = 2[-(1(-1))] = 2$$

easy

$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos(x)}{\tan(x)\sec(x)}\,\mathrm{d}x$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos(x)}{\frac{\sin(x)}{\cos(x)}\frac{1}{\cos(x)}}\,\mathrm{d}x = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos(x)}{\frac{\sin(x)}{\cos^2(x)}}\,\mathrm{d}x = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos(x)\frac{\cos^2(x)}{\sin(x)}\,\mathrm{d}x = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos(x)\frac{1-\sin^2(x)}{\sin(x)}\,\mathrm{d}x,$$

$$u = \sin(x), \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = \cos(x), \qquad \mathrm{d}x = \frac{1}{\cos(x)}\,\mathrm{d}u,$$

$$\int_{\frac{\sqrt{2}}{2}}^{1} \cos(x)\frac{1}{\cos(x)}\frac{1-u^2}{u}\,\mathrm{d}u = \int_{\frac{\sqrt{2}}{2}}^{1} \frac{1-u^2}{u}\,\mathrm{d}u = \int_{\frac{\sqrt{2}}{2}}^{1} \frac{1}{u}\,\mathrm{d}u - \int_{\frac{\sqrt{2}}{2}}^{1} u\,\mathrm{d}u = \left[\ln|u| - \frac{u^2}{2}\right]_{\frac{\sqrt{2}}{2}}^{1} =$$

$$= -\frac{1}{2} - \left(\ln\left(\frac{\sqrt{2}}{2}\right) - \frac{\frac{1}{2}}{2}\right) = -\frac{1}{2} - \ln\left(\frac{\sqrt{2}}{2}\right) + \frac{1}{4} = -\frac{1}{4} - \ln\left(\frac{\sqrt{2}}{2}\right)$$

2026 14th April

derivative

easy

Find the slope $\left(y-x^5\right)^2 = x\left(1+x^2\right)^2$ at the $(1,3)$ point.

$$\partial_x = 2(y-x^5)(-5x^4) = \left(1+x^2\right)^2 + 2(1+x^2)(2x) \implies$$
$$\implies 2(y-x^5)(-5x^4) - \left(1+x^2\right)^2 - 2(1+x^2)(2x) = 0$$
$$\partial_y = 2(y-x^5) = 0$$
$$\partial_x f(a) = 2(3-1)(-5) - (1+1)^2 - 2(1+1)(2) = -20-4-8 = -32$$
$$\partial_y f(a) = 2(3-1) = 4$$
$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{\partial_x f(a)}{\partial_y f(a)} = -\frac{-32}{4} = 8$$

medium

Let $f(x) = \sin\left(\frac{e^{\tan(x)}}{\ln(x)}\right)$, find $f'(e)$

$$f'(x) = \cos\left(\frac{e^{\tan(x)}}{\ln(x)}\right)\left(\frac{\sec^2(x)e^{\tan(x)}\ln(x) - \frac{e^{\tan(x)}}{x}}{\ln^2(x)}\right) = \cos\left(\frac{e^{\tan(x)}}{\ln(x)}\right)\left(\frac{e^{\tan(x)}\left(\sec^2(x)\ln(x) - \frac{1}{x}\right)}{\ln^2(x)}\right),$$
$$f'(e) = \cos\left(e^{\tan(e)}\right)e^{\tan(e)}\left(\sec^2(e) - \frac{1}{e}\right)$$

integral

beginner

$\int_{\frac{5\pi}{4}}^{\frac{9\pi}{4}} e^{\sin(x)}e^{\cos(x)}(\cos(x)-\sin(x))\,\mathrm{d}x$

$$\int_{\frac{5\pi}{4}}^{\frac{9\pi}{4}} e^{\sin(x)+\cos(x)}(\cos(x)-\sin(x))\,\mathrm{d}x,$$
$$u = \sin(x)+\cos(x), \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = \cos(x)-\sin(x), \qquad \mathrm{d}x = \frac{1}{\cos(x)-\sin(x)},$$
$$\sin\left(\frac{5\pi}{4}\right) + \cos\left(\frac{5\pi}{4}\right) = -\frac{\sqrt{2}}{2} + -\frac{\sqrt{2}}{2} = -\sqrt{2}$$
$$\sin\left(\frac{9\pi}{4}\right) + \cos\left(\frac{9\pi}{4}\right) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2}$$
$$\int_{-\sqrt{2}}^{\sqrt{2}} e^u(\cos(x)-\sin(x))\frac{1}{\cos(x)-\sin(x)}\,\mathrm{d}u = \int_{-\sqrt{2}}^{\sqrt{2}} e^u\,\mathrm{d}u = \left[e^u\right]_{-\sqrt{2}}^{\sqrt{2}} = e^{\sqrt{2}} - e^{-\sqrt{2}}$$

easy

$\int_0^\pi x\cos^2(x)\,\mathrm{d}x$

	D	I
$+$	x	$\cos^2(x)$
$-$	1	$\frac{x}{2}+\frac{\sin(2x)}{4}$
$+$	0	$\frac{x^2}{4}-\frac{\cos(2x)}{8}$

$$\int_0^\pi x\cos^2(x)\,\mathrm{d}x = \left[x\left(\frac{x}{2}+\frac{\sin(2x)}{4}\right)-\left(\frac{x^2}{4}-\frac{\cos(2x)}{8}\right)\right]_0^\pi = \frac{\pi^2}{2}-\frac{\pi^2}{4}+\frac{1}{8}-\frac{1}{8}=\frac{\pi^2}{4}$$

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derivative

easy

Let $f(1) = 1$ and $f'(1) = 3$, find the derivative of $f(f(f(x))) + (f(x))^2$ at $x = 1$

$$f(x) = x^3, \qquad f'(x) = 3x^2,$$

$$f(f(f(x))) + (f(x))^2 = \left(\left(x^3\right)^3\right)^3 + \left(x^3\right)^2 = x^{27} + x^6,$$

$$\frac{\mathrm{d}}{\mathrm{d}x}(f(f(f(x))) + (f(x))^2) = 27x^{26} + 6x^5 \overset{1}{\Longrightarrow} = 27 + 6 = 33$$

medium

Let $f(x) = \sum_{n=1}^{50} nx^n$, find $f'(1)$

$$f(x) = x + 2x^2 + 3x^3 + ... + 50x^{50}$$

$$f'(x) = \sum_{n=1}^{50} n^2 x^{n-1} = 1 + 4x + 9x^2 + ... + 2500x^{49}$$

$$f'(1) = \sum_{n=1}^{50} n^2 = \frac{50 \cdot 51 \cdot 101}{6} = 42925$$

integral

beginner

$\int_0^{\frac{\pi}{2}} \frac{1}{\sec(x) + \tan(x)} \, \mathrm{d}x$

$$\int_0^{\frac{\pi}{2}} \frac{1}{\frac{1}{\cos(x)} + \frac{\sin(x)}{\cos(x)}} \, \mathrm{d}x = \int_0^{\frac{\pi}{2}} \frac{1}{\frac{1+\sin(x)}{\cos(x)}} \, \mathrm{d}x = \int_0^{\frac{\pi}{2}} \frac{\cos(x)}{1 + \sin(x)} \, \mathrm{d}x,$$

$$u = \sin(x), \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = \cos(x), \qquad \mathrm{d}x = \frac{1}{\cos(x)} \, \mathrm{d}u,$$

$$\int_0^1 \frac{1}{1+u} \, \mathrm{d}u = [\ln|1+u|]_0^1 = \ln(2)$$

easy

$\int_0^1 \frac{\arctan(x) - x\arctan(x)}{1 - x + x^2 - x^3} \, \mathrm{d}x$

$$1 - x + x^2 - x^3 = (1-x) + x^2(1-x) = (1-x)(1+x^2)$$

$$\int_0^1 \frac{\arctan(x) - x\arctan(x)}{1 - x + x^2 - x^3} \, \mathrm{d}x = \int_0^1 \frac{\arctan(x)(1-x)}{(1-x)(1+x^2)} \, \mathrm{d}x = \int_0^1 \frac{\arctan(x)}{1+x^2} \, \mathrm{d}x =$$

$$= \frac{1}{2}\left[\arctan^2(x)\right]_0^1 = \frac{\arctan^2(1)}{2}$$

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derivative

easy

medium

integral

beginner

easy